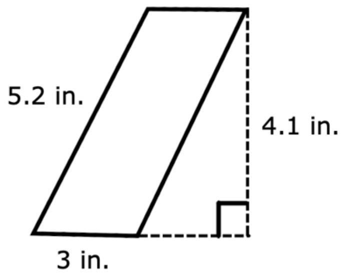
Grade 6 Accelerated	Name___Answer Key_______
Unit 7	Date____________________
Lesson 4 Homework	Period___________________

A parallelogram with dimensions labeled is shown below for questions 1-2.



1. Which dimension is not needed to find the area of the parallelogram? Explain.
5.2 is not needed because it represents the side length or slant height but not the height of the parallelogram.

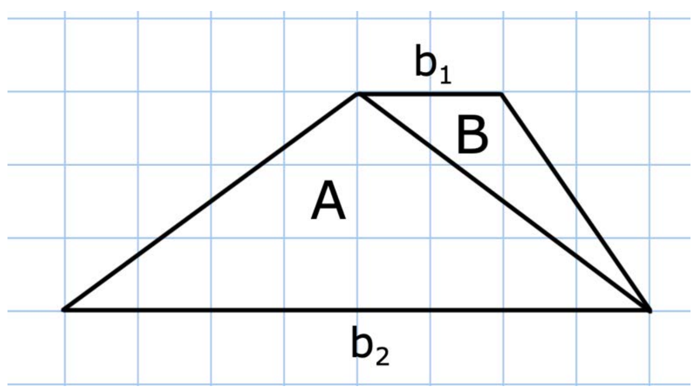
2. Fill in the blanks to find the area of the parallelogram.

Area of parallelogram = $b \cdot h$

Area = __3__ · _4.1__

Area = _12.3_ in.^{2}

A trapezoid can be decomposed into two triangles with the same height. Use the diagram below for questions 3-6.



3. The base of triangle A, b_2, is __8___ units and its corresponding height is ___3___ units. Therefore, the area or triangle A is ___12_____.
$\left(\frac{1}{2} \cdot 3 \cdot 8\right)$

4. The base of triangle B, b_1, is __2___ units and its corresponding height is ___3___ units. Therefore, the area or triangle B is ___3_____.
$\left(\frac{1}{2} \cdot 3 \cdot 2\right)$

5. The area of the trapezoid is __$12 + 3 = 15$__ units^2.

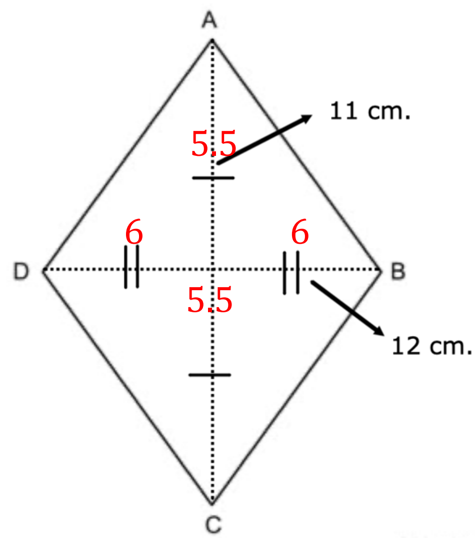
6. Use the formula for area of a trapezoid to verify your solution is correct.

Area = $\frac{1}{2}h(b_1 + b_2)$.

Area = $\frac{1}{2}$ __3__ (__2__+ __8__)

Area = __15__ units^2

Rhombus $ABCD$ is shown. Diagonal BD is 12 centimeters and diagonal AC is 11 centimeters. Use the diagram below for questions 7-10.



7. The base of triangle ABC is __11__ cm. and its corresponding height is __6__ units. Therefore, the area of triangle ADC is __33__.
$\left(\frac{1}{2} \cdot 6 \cdot 11\right)$

8. The base of triangle ADC is __11__ cm. and its corresponding height is __6__ units. Therefore, the area or triangle ADC is __33__.
$\left(\frac{1}{2} \cdot 6 \cdot 11\right)$

9. The area of the rhombus is __$33 + 33 = 66$__ cm^2.

10. Use the formula for area of a rhombus to verify your solution is correct.

Area = $\frac{1}{2}(d_1)(d_2)$.

Area = $\frac{1}{2}$(__12__)(__11__)

Area = __66__ units^2